Mathematics - Graduate Level Assessment 18

Instructions

- Answer all questions.

Multiple Choice

Question 1

(2 marks)

Which situation could be represented by the expression 6 x 2?

A. Rocco read six books in two months.

B. Rocco hiked six miles each day for two days.

C. Rocco spent six hours on homework for two days.

D. Rocco has six siblings and two of them are girls.

E. Rocco had a total of six tennis balls in two cans.

Question 2

(2 marks)

Find the interval in which the smallest positive root of the following equations lies: $x^3 - x - 4 = 0$. Determine the roots correct to two decimal places using the bisection method

A. 2.5

B. 1.7

C. 1.9

D. 2.2

E. 1.8

Question 3

(2 marks)

Find the generator for the finite field Z_7.

A. 1

B. 8

C. 0

D. 7

E. 5

Question 4 *(2 marks)*

A company makes 5 blue cars for every 3 white cars it makes. If the company makes 15 white cars in one day, how many blue cars will it make?

A. 25

B. 35

C. 17

D. 13

E. 9

Question 5 *(2 marks)*

A worker on an assembly line takes 7 hours to produce 22 parts. At that rate how many parts can she produce in 35 hours?

A. 110 parts

B. 245 parts

C. 770 parts

D. 220 parts

E. 140 parts

True / False

Question 1

(2 marks)

The correct answer to 'Find the sum of $\Sigma_{(n=1)}^{(\infty)} 2/n^2 + 4n + 3$' is '0.8333'.

- **True**
- **False**

Question 2

(2 marks)

In a group, the equation $(ab)^{(-2)} = b^{(-2)}a^{(-2)}$ is false for all groups, and $(ab)^n = a^nb^n$ is also false for all groups.

- **True**
- **False**

Question 3

(2 marks)

The estimated sum of 153 and 44 falls within the range of 400 and 599.

- **True**
- **False**

Question 4

(2 marks)

The value of the integral $\int_2^4 \frac{\sqrt{\log(9-x)}}{\sqrt{\log(9-x)}+\sqrt{\log(x+3)}}\,dx$ is 2.2.

- **True**
- **False**

Question 5

(2 marks)

The correct answer to 'Compute $\int_{(|z| = 1)} z^2 \sin(1/z)\,dz$. The answer is Ai with i denoting the imaginary unit, what is A?' is '-0.5'.

True

False

Long Form

Question 1 *(10 marks)*

How many two-digit prime numbers have a units digit of 7?

Type your answer here...

Question 2 *(10 marks)*

How many paths are there from A to B on the lines of the grid shown, if every step must be up or to the right?[asy]size(3 cm,3 cm);int w=6;int h=3;int i;for (i=0; i

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